EFAZ HOSSAIN

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PROFESSIONAL SUMMARY

Data Scientist / ML Engineer with experience building end-to-end data pipelines, predictive models, and production-ready analytics systems. Skilled in Python, SQL, and cloud-based workflows (AWS/Azure), with hands-on experience in ETL pipelines, machine learning (Scikit-learn, TensorFlow), and data warehousing (dbt, PostgreSQL). Proven ability to deliver business-impacting insights through predictive modeling (ROC-AUC 0.846) and scalable data systems processing 10K+ events.

EDUCATION

The University of Texas at Arlington **Arlington, TX**

Bachelor of Science in Data Science (Biology Concentration) August 2020 – December 2025

TECHNICAL SKILLS

- **Languages & ML:** Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL
- **Data Engineering:** ETL/ELT, dbt, PostgreSQL, Data Modeling, Data Warehousing, S3/MinIO
- **Analytics & Visualization:** Tableau, Power BI, Matplotlib, Seaborn, KPI Development
- **Cloud & Tools:** AWS, Azure, Docker, Git, Linux, Bash

EXPERIENCE

Research Assistant
The University of Texas at Arlington — Arlington, TX
May 2025 – Dec 2025

- Developed a novel gradient-free optimization algorithm (SQH) for survival models with competing risks, improving model estimation without reliance on gradient-based methods
- Designed and executed simulation studies benchmarking SQH vs. PNCG and EM, evaluating estimation accuracy and computational efficiency across multiple scenarios
- Applied survival analysis and statistical modeling to real-world melanoma datasets, validating model performance and interpretability in a healthcare analytics context
- Built R-based data pipelines for synthetic data generation, model training, and performance evaluation, enabling reproducible experimentation
- Collaborated on a peer-reviewed research manuscript, contributing to methodology development and experimental validation

PROJECTS

Multimodal Churn Risk Detection | Python, Scikit-learn, TensorFlow, SHAP

- Built an explainable multimodal churn intelligence system combining structured customer data with synthetic support-note text, achieving 0.847 ROC-AUC and 0.654 PR-AUC.
- Engineered a reproducible ML pipeline for feature generation, text embeddings, model training, evaluation, and reporting.
- Compared Logistic Regression and XGBoost models, selecting Logistic Regression for stronger recall and 2.85 lift among the top 10% highest-risk customers.
- Developed SQL-based churn segmentation analyses to identify high-risk groups by contract type, tenure, payment method, and revenue exposure.
- Implemented SHAP explainability and a Streamlit app for customer-level churn scoring, driver analysis, and retention decision support.

Agentic DataOps Platform | Docker, PostgreSQL, dbt, MinIO

- Built a containerized data platform integrating object storage with a data warehouse
- Developed ETL/ELT pipelines (dbt) with raw → staging → mart transformations
- Processed 10K+ events across partitioned ingestion workflows and implemented data quality checks and freshness monitoring
- Built automated remediation workflows to improve pipeline reliability

Clinical Survival Analysis |Pandas, Lifelines, scikit-survival, scikit-learn, Matplotlib

- Built a reproducible breast cancer survival analysis pipeline using METABRIC clinical data, applying Kaplan-Meier curves, log-rank tests, Cox proportional hazards modeling, and Random Survival Forests.
- Identified statistically significant survival differences by tumor stage (log-rank $p = 9.35e-27$) and ER status ($p = 0.035$), with visual survival curves for clinical interpretation.
- Trained a Random Survival Forest model achieving a 0.681 test C-index and analyzed survival risk drivers using Cox hazard ratios and permutation-based feature importance.